International Journal of Medical Informatics 161 (2022) 104711

Contents lists available at ScienceDirect

International Journal of Medical Informatics

journal homepage: www.elsevier.com/locate/ijmedinf

Perspectives on record-keeping practices in MDT meetings and meeting record utility

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ARTICLE INFO

Keywords:
MDT records
Meeting records
Multidisciplinary Team Meetings

ABSTRACT

Background and purpose: Working collaboratively as a multidisciplinary team in the treatment and care of cancer patients is proven effective in increasing the quality of patient care. A multidisciplinary team (MDT) meeting (MDTM) is the main vehicle that facilitates this collaborative work between different healthcare specialities, and an appropriate meeting record is essential to communicate the meeting's outcomes. There is limited research to date regarding MDTM documentation, and here we report on a sample of healthcare professional's perspectives on current practices.

Methodology: A survey, distributed to a purposive snowball sample, is used to collect the perceptions on record-keeping at MDTMs from involved healthcare workers. The survey is descriptive and exploratory in nature and uses closed and open-ended questions offered in both English and Swedish.

Results: With a response of 37 healthcare workers, several commonly understood practices of MDTMs are confirmed, documentation is mostly electronic, encompasses suggested information, and the record is mostly acceptable in quality. The issues of responsibility, registering attendance, and verification of documentation can be improved.

Conclusion: Electronic documentation is a laudable step that shows advancement in MDTM record-keeping. The highlighted quality of the records suggests that MDTM proceedings are reasonably well documented. There remain some important questions, with regard to standardization, centralization, and the responsibility for record-keeping at MDTMs.

1. Introduction

Multidisciplinary team (MDT) meetings (MDTMs) have become an essential part of cancer care, worldwide, and there is a broad general agreement on the way that MDTMs work. MDTMs are most often held weekly, and some MDTs meet more or less frequently [1]. MDTs are linked to several benefits to patients and to the healthcare system [2,3]. Since the various specialists work collaboratively on each patient case, the richness of their expert knowledge and experience combined with all necessary patient information leads to more accurate diagnosis and disease staging [3], and better treatment recommendations [4]. MDTs ensure timely treatment for patients and that patients get care from appropriately skilled professionals [5].

General surgeons are one of the specialties that wish for a cancer care medical record that provides consolidated patient information electronically as well as more efficient means to communicate test and treatment results [6]. Since MDTMs are an important co-ordination mechanism in patient care, MDTM records should include details of MDTM proceedings (meta-data) and support MDT activities that relate to patient treatment and care [7,8]; accuracy, completeness, accessibility, and usability are paramount. The question of whether comprehensive records of MDTM proceedings are now in place still requires exploration as little is reported about record-keeping at MDTMs, although the adoption of methods to support real-time multimedia record-keeping at MDTMs is suggested [9], and it is over 20 years since a technologically supported meeting space was proposed [10].

We present health professionals perspectives on their current record-keeping practices. We enquire if electronic records are in place that detail MDTM proceedings, the type of information that is included in the record, and the record utility to effectively assist the follow-on processes

Abbreviations: MDT, Multidisciplinary team; MDTM, Multidisciplinary team meeting; PCD, Patient Case Discussion; CPG, Clinical Practice Guideline; SCP, Standardised Care Process.

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https://doi.org/10.1016/j.ijmedinf.2022.104711
Received 6 July 2021; Received in revised form 17 January 2022; Accepted 27 January 2022
Available online 10 February 2022
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post-MDTM. It is believed that electronic record-keeping is a better aid than paper record-keeping [11,12]. The study is conducted in Sweden, which has, for the most part, an electronic system of gathering information in healthcare processes for sharing and communication [13]. Electronic medical records (EMRs) enable health care providers to improve patient safety and health care quality, reduce medical errors, improve communications between clinicians, and increase access to medical knowledge and patient histories [14].

2. The MDTM and its Record

Multidisciplinary medical teams (MDTs) comprise all the specialities that are involved in the care of a patient and are in evidence in several types of clinical settings. MDTs are most widely used in providing services to cancer patients where integration and co-ordination of services is critically important. Typically cancer patients develop or already have co-morbidities, and several specialities are involved in their care, which makes medical decision and interventions difficult and prone to errors [15]. The MDT meeting (MDTM) can be described as a collaborative forum where the various specialities share and discuss data and information on the patient that arise from the independent investigations, such as radiology, pathology, endoscopy, patient symptoms and clinical observations [16]. Indeed, through this collaboration new knowledge is generated that provides a more comprehensive view of the patient for the MDT, and helps refine the diagnosis and treatment recommendations for the individual patient [17].

Clinical practice guidelines for several types of cancer advise that MDTs be used in agreeing the patient's diagnosis and guiding the treatment plan for the patient [18]. Such MDTs are considered to improve patient outcomes and patient safety; and lower the risks associated with healthcare processes [19,20]. MDTs are also shown to improve planning of therapy, adherence to recommended preoperative assessment, pain control, and adherence to medications [21].

MDTMs are preceded by a number of pre-MDTM activities that inform the MDTM and facilitate comprehensive discussion [22,23]. MDTMs discuss each patient case in a structured way; MDTM discussions and decisions, in turn, influence the events and activities that take place post-MDTM [22]. The patient status, clinical information, endoscopic, radiology and pathological findings are presented in order to agree the diagnosis and disease stage; a discussion follows on the most appropriate proposal for the treatment or management plan [7,24], taking into consideration the co-morbidities and patient preferences wherever possible. The availability of high-quality, comprehensive information is needed for MDTs to make clinical decisions; other factors such as the meeting structure, position of case in the agenda, case preparation and case load also impact [25].

Significant resources are consumed in MDTMs and related work processes [26], and thus their value in healthcare is scrutinized [27]. The collaborative work of MDTs not only involves MDTMs, but many related tasks that happen pre- and post-MDTMs, with time and financial costs attached. MDTMs are now considered essential to cancer care; and the associated processes should be optimized and lead to improved patient outcomes [27]. Though MDTMs have primarily focused on findings from pathology, radiology, and surgery [28], the knowledge of patients' co-morbid conditions, current health status and preferences also assist in making the most appropriate decision on treatment options [29]. Patient-centred information is presented at MDTMs by MDT members who have examined the patients and thus meeting management is an essential factor to successful decision-making [27]. Although the MDT and MDTM that we describe here can be found in hospital and cancer care settings, the processes and findings apply to any team where different specialists co-operate and collaborate, share information and make decisions on a shared goal.

The decisions made at the MDTM determine the processes that follow, and clear instructions are required on the tasks to be conducted post-meeting [7]. The level of organisation and type of communication in MDTMs influence patient care quality [29] and there is a need for a technological solution to facilitate an information-rich record to address concerns such as misinterpretation and contradictions from the lack of a formal, comprehensive record [22]. Though some clinicians may not require access to MDTM records, having original data on record would facilitate recall and reflection when needed, and having the record in electronic form yields better access [22]."With robust EMRs, both the patient and providers will have ready access to critical medical information, information which can be updated continuously and shared across providers" [14, p.139].

MDTs are interested in information such as 'who was there?', 'who participated in a discussion?', 'which data were discussed?', 'what is Dr. X's opinion?', and 'were differential diagnoses considered?' [30]. It is necessary to keep a record of the agreed diagnosis, the disease stage, and the recommended treatment, along with the rationale for the decision. It is also important to record any disagreements that may arise during a discussion and potentially discuss any alternative suggestions on a given matter with the patient [25], as more than one reasonable option may be raised concerning the management of an individual patient case [29].

Information organisation in health records is important as poor organisation increases the chance of error [11]. The use of symbols and acronyms should be clear. The use of symbols in medical documentation, particularly within and between medical teams, is a safety issue due to ambiguity because the symbols are largely unknown to medical professionals [31]. The use of abbreviations is discouraged as they too might be ambiguous [32], and lead to misinterpretation [22].

Not only do MDTMs provide an opportunity for communication between various specialists, but MDTMs provide learning opportunities [29]. "Records of the case discussions …provide a valuable learning repository for ongoing training and development" [22, p.462].

3. Methodology

In this cross-sectional study, we utilize an online questionnaire to capture perspectives of medical professionals who work in, or closely with, hospital oncology departments and are involved in patients' multidisciplinary care. We target both MDTM attendees and medical professionals who do not attend MDTMs but utilize MDTM records during patient care.

Purposive sampling is used to reach participants through contacts used from regional health bodies, hospitals, and associations linked with medical professionals. Ethical and GDPR guidelines are observed regarding data processing and storage. No personal or identifying information were gathered.

The questionnaire was developed on the Survey & Report tool with the questions based on earlier surveys [33,34] and on extant literature addressing the need for an MDTM record [35,36], information to be included in the record [30,29,25], and the importance of keeping a structured record [30]. The questionnaire was piloted with two healthcare professionals before distribution.

The questionnaire, available in English and Swedish, contains 5-point Likert-scales and open-ended questions. Responses were analyzed independently by both authors, and any differences in interpretation discussed. Thematic analysis is used for text responses.

When we refer to record-keeping and documentation, these tasks are analogous to what is commonly known as 'minute taking'. In the case of the MDT meeting, the account of the individual patient case discussions needs to be documented in the individual's electronic record, as well as having a record of the MDT meeting itself.

This study is approved and registered with the Ethics Committee for Karlstad University. The registration reference number is HS 2019/1344. A post-graduate student award utilized early results of the survey [37].

4. Results

A total of 37 participants completed the survey, working in 14 hospitals, across 8 regions in Sweden. 36 of 37 participants work in teams that attend MDTMs, mostly weekly, and 27 (73%) of participants' institutions hold MDTMs over video links, 59% of which are weekly. The 37 participants comprise 14 nurses, 7 physicians, 6 surgeons, 3 oncologists, 2 radiologists, 1 pathologist, 2 administrators, and 2 are medical secretaries.

Participants perform various tasks before, during, and after MDTMs as summarised in Table 1. Several of these roles are manual. Data for each patient are collected from various sources such as radiology and pathology. The tasks conducted 'During MDTM' are performed concurrently with the discussion.

Tasks highlighted by participants aid the flow of discussions and follow-on processes. Before MDTMs, tasks comprise of collecting data necessary, scheduling tasks, and preparing for discussion. During MDTMs are tasks such as controlling the meeting flow, operating equipment, and taking personal notes relevant to the note-takers role responsibilities. After MDTMs, tasks include documentation, communicating with patients and contact nurses, sending referrals, and scheduling further investigations or surgery.

The top three reasons selected for attendance in MDTMs, from 36 participants (in Table 2), are i) to participate and contribute to discussions (26, 72.2%), ii) to present a patient case (18, 50%), and iii) to get information for coordination of care (15, 41.7%). While iv) getting information for treatment and care, and v) learning, are reasons to attend the meeting for 10 (27.8%), respectively. The majority of MDTMs are formally documented as soon as the meeting is over, and no later than 72h afterwards. During MDTMs, some form-filling and other paper records are completed (7, 20%), while one (2.9%) reports that paper recording happens after the MDTM. Four (11.5%) report that audio or audio visual recordings are taken at their MDTMs. Eighteen (51.4%) report that text is recorded electronically, and 21 (60%) take personal notes.

As shown in Table 3, the responsibility for documenting MDTM proceedings and decisions lies mainly with the physician who refers the patient case to the MDT (called here 'referring physician'). Medical secretaries, designated data managers, coordinators, nurses, administrators, quality register secretaries, and any role that will treat the patient afterwards such as surgeons, doctors familiar with the patient case, contact nurses, or other team members are all counted among those who are sometimes responsible for documentation (Table 3). Additional survey comments highlight that responsibility for documentation is determined on a case-by-case basis; one of the team members may be delegated in advance as the person responsible for documentation, or the doctor who reported the patient case, or the appointed surgeon who will handle the case, or/and the MDTM chairperson, may assume the responsibility for the discussion record. Further explanations added by 8 participants show an agreement that the notes that are taken at the meetings are later entered into the electronic medical system by the clinician responsible, or through the dictation of such notes to another who will enter the information into the electronic medical system.

MDTM attendees may or may not be noted; 30 (of 36) who attend MDTMs affirm that meeting attendees are registered as attendees, 5 responded 'no', and 1 does not know if attendees are recorded. One

Table 2
Reasons for attendance at MDTMs

Reasons of attendance	Participants (%)
Participate/contribute to a discussion	26 (72.2%)
Present a case for discussion	18 (50%)
Get information for coordination of care	15 (41.7%)
Get information for treatment and care planning	10 (27.8%)
Learn / Education	10 (27.8%)
Administrative/technical support reasons	4 (11.1%)
Other	1 (2.8%)
Total Participants	36 (100%)

Participants were free to select more than one choice.

Table 3
Persons identified that ideally should be responsible for documenting MDTM proceedings/ decisions.

Responsible person	Participants (%)
Attending (Referring) physician	16 (50%)
Medical secretary	8 (25%)
Designated data manager	6 (18.8%)
Each person who will treat the patient takes independent notes for their own use	6 (18.8%)
Coordinator	5 (15.6%)
Nurses	4 (12.5%)
Other	5 (15.6%)
Administrator	2 (6.3%)
Total Participants	32 (100%)

Table 1
Tasks and roles of health professionals (survey participants) before, during, and after MDTMs.

Pre-MDT meeting	During MDT meeting	Post-MDT meeting
Plan dates for MDTMs	Start up equipment	Circulate attendance list
Announce cancellation (if MDTMs cancelled)	Record Attendance	Record notes & ensure availability for admitted patients
Prepare for MDTM:	Chairman:	Document suggestions & discussion in medical record
· Accept case registration	· Facilitator	Ensure referrals prepared and sent
· Schedule cases	· Maintain order	Ensure SCP[1] measuring points are entered in the *Journalen*,
· Prepare case list	· Ensure Diagnosis & Stage agreed	the Swedish patient accessible online electronic medical record
· Compile MDTM signups & any additions	· Take any rebooking request	
· Distribute Agenda		
Assign contact nurse for each patient	Ensure steps e.g. formal staging completed	'Document cases I presented'
Prepare patient case (including radiology/ pathology demonstrations)	Gather information to coordinate care	Plan treatment
		Administer or excuse conclusions from MDTM
Read cases listed		
Advise patient their case is listed for MDTM & check consent	Present how the patient feels	Dictate Notes
		Communicate with contact nurse & patient
Prepare for discussion:	Present Patient Cases	Code in medical record that MDTM completed
· Define clear question to be addressed	Note referrals needed	Order/Schedule additional exams/tests
· Summarise medical history	Note examinations to order	Order/Schedule surgery
· Ensure patients listed are fully examined, including pathology, x-rays, PET-CT & Scintigraphy	Note bookings to be made for surgery	Schedule return visits
	Follow-up decisions	Discuss decisions & present proposed treatment plan to patient
· Ensure all necessary information is available	Dictate decisions	Document patient's receipt of proposal

[1] SCP - Standardised Care Process

Table 4
Comments on the people responsible for record-keeping at MDTMs.

Role	Comment
Physician	The doctor appointed to present the patient case takes notes and then dictates them (can be one per clinic if several are dealing with the patient). In addition, both nurses and secretaries make their own notes for their own role benefit. The doctor responsible for the care process for that MDTM applies; the same doctor leads MDTM. The responsible doctor dictates to the responsible secretary. The doctor who signed up the patient is responsible for the presentation of the case to the MDTM is the one who documents. If the doctor signing up is prevented from attending, it is their deputy who ensures the case is documented.
Oncologist	"We document for patients already recorded at the oncologist's [clinic] (e.g. we registered) or on patients who will come to us".
Surgeon	The surgeon who is appointed to present that particular patient. "I am not familiar with how surgeons and oncologists document".
Chairman	The clinical chair is responsible for keeping discussions and decisions recorded. Sometimes referrals and tasks are delegated to others by the chairman. Doctors appointed within MDTM, may chair the meeting but also other participating doctors who are familiar with the case. Determined in connection with each patient case. Template is missing.
Nurse	Contact nurse
Delegated	We have a doctor responsible for the entire MDTM who delegates record keeping to other participants. The doctors who attend MDTM decide in advance the individual responsible for documenting each patient's case. A doctor's rapporteur, depending on the decision. When surgery is decided records are dictated by surgeons on the MDTM decision. If [decision] is oncological treatment, the oncologist writes the medical record. Doctors are assigned different patients and document in the medical record after the meeting.
Medical Secretary	The medical secretary documents directly in the medical record for the chairman on site who is either an oncologist or a urologist.

participant adds that attendees are noted selectively; doctors always by name, nurses sometimes by name or more often by function. In other cases, there is no system in place to register attendees and in national MDTMs registration is said to be a clear shortcoming.

Table 4 presents comments made in response to 'Who is responsible for documenting MDK/MDTM proceedings/decisions?'. The column 'Role' refers to the roles that the survey respondent indicated is responsible. Typically the role of the survey participant differs from the role they indicated. For example, the nurse is identified as the role that should be responsible for making the record by several, but nurses appear to point to medical secretaries and physicians as the roles responsible for entering the information in the record.

The survey enquires about health professionals' opinion on the adequacy of information contributed by various healthcare professionals (clinical, radiology, oncology). The responses are broadly very positive. In patient case discussions (PCDs), the clinical information is considered sufficient (29, 80.5%), and similarly, the radiology data presented (28, 77.8%) is considered sufficient. Information on pathology is not considered so satisfactory (19, 54.3%) nor information on the patient's co-morbidities (17, 47.3%). Although reported by 15 (41.7%) that patient preferences are available during PCDs, 7 (19.5%) disagree. The patient's preference is always taken into account according to 21 (58.4%), and 5 (13.9%) disagree.

Twenty-eight (77.8%) respondents report that the diagnosis is always agreed and that there is consensus on the disease stage (29, 80.5%). Consensus on the management plan is always reached according to 30 (83.3%). These strongly positive results are slightly marred by the higher numbers of "undecided" responses and disagreements. Five "undecided" responses account for 13.9% in both the question of consensus on disease stage, and consensus on management. While participants report that lack of consensus is rare, measures are in place for resolving any differences in opinion.

When a patient case needs to be discussed more than once, it is more likely to be due to missing or inadequate information (25, 69.4%), new complications arising in the patient condition (22, 61.1%), or because patients need to be discussed at different stages in the care pathway (20, 55.6%) than because of patient disagreement with recommendations (5, 13.9%) or the need for a follow-up review (4, 11.1%).

The management or treatment recommendation agreed is mostly (22, 61.1%) or completely (13, 36.1%) based on clinical guidelines (35, 97.2%) and mostly (24, 68.6%) or completely (10, 28.6%) based on clinical data (34, 97.1%). Co-morbidities are also considered during the discussions (30, 83.4%) and 23 (63.9%) of participants indicate that patient preferences play a role in reaching the agreement on treatment recommendations. Detailed results presented in Table 5 suggest that there could be better alignment between clinical practice guidelines (CPGs) and patient preferences in the MDT recommendations. Speculation on reasons for the misalignment is not made here, since the result might be explained by several hypothesis that need to be explored in further research.

The majority (31, 86.1%) agree they have clear documentation of MDTM decisions. While 22 (61.1%) agree the rationale behind the decisions is clearly documented, fewer (17, 47.3%) agree that discussions leading to the decisions are noted. Most (25, 89.3%) believe the quality of MDTM record-keeping is adequate, while slightly less (20, 74.1%) believe that MDTM documentation adequately guides the flow of follow-on activities. A minority regard the MDTMs as less than adequate.

Less than half surveyed (16, 44.5%) believe that treatment options are clearly documented, and others either 'disagree' or remain neutral on whether treatment options are clearly documented. Perceptions around the verification of the records show some uncertainty, as demonstrated in the responses in Table 6 which reports on statements regarding MDTM documentation.

The agreed management plan is usually followed (24/28, 86%) and guides the flow of follow-on processes (20, 74%). Typically, deviations from decisions are justified (18, 64%) and clearly documented (24, 86%). While MDTM records are referenced during patient care (29/34, 85%), 5 (15%) do not report the need.

The survey asks participant's perceptions of the general MDTM organisation where they work. Overall, the majority say the MDTMs work well, though there is a need to re-examine the technical challenges being experienced with equipment. Participants' comments are presented in Table 7.

5. Discussion

The core MDT participants, namely, physicians, nurses, oncologists, pathologists, surgeons, radiologists, and administrators, are represented in this study and the single largest group of respondents are nurses. The use of open-ended questions enabled the collection of further information from our participants in free text.

5.1. The ability to reach consensus in a patient case discussion will aid clarity of documentation

The health professionals' view, represented in this study, is that the information required to aid the MDTM discussions is sufficient in most patient cases; diagnosis and disease staging are usually agreed, and consensus is reached on the treatment recommendation. The fact that the discussions are considered so satisfactory is an important consideration in respect of well-documented MDTMs, as resulting documentation can be expected to lead to clearer guidance on the steps to follow. Some study participants maintain that they have had little to no exposure to cases where consensus is not reached and that for the MDTM not to reach consensus would be a rare event. Respondents explain that cases for which consensus is not reached are remedied by i) applying different options such as re-discussion of a patient case following additional tests or ii) retrieval of information on patient preferences where

Table 5
Inputs to MDT decision on management and treatment recommendations.

To what extent is the management/ treatment recommendation agreed based on …	Not at all	To a lesser extent	Somewhat	To a great extent	Completely	Don't know
	Frequency (%)					
Clinical guidelines	0/36 (0%)	0/36 (0%)	0/36 (0%)	22/36 (61.1%)	13/36 (36.1%)	1/36 (2.8%)
Patient preferences	0/36 (0%)	4/36 (11.1%)	7/36 (19.4%)	20/36 (55.6%)	3/36 (8.3%)	2/36 (5.6%)
Clinical data	0/35 (0%)	0/35 (0%)	0/35 (0%)	24/35 (68.6%)	10/35 (28.6%)	1/35 (2.9%)
To what extent are co-morbidities considered in the discussions during the MDTM	0/36 (0%)	3/36 (8.3%)	3/36 (8.3%)	19/36 (52.8%)	11/36 (30.6%)	0/36 (0%)

Table 6
Perceptions on the clarity and accuracy of the MDTM record.

	Strongly Disagree	Disagree	Neither Agree nor Disagree	Agree	Strongly Agree
			Frequency (%)		
Decisions taken at an MDTM are always clearly documented	0/36 (0%)	3/36 (8.3%)	2/36 (5.6%)	17/36 (47.2%)	14/36 (38.9%)
The rationale behind the decisions taken is always clearly documented	0/36 (0%)	5/36 (13.9%)	9/36 (25%)	19/36 (52.8%)	3/36 (8.3%)
Discussions leading to the decisions are always clearly documented	1/36 (2.8%)	9/36 (25%)	9/36 (25%)	15/36 (41.7%)	2/36 (5.6%)
Various treatment options considered are always clearly documented	0/36 (0%)	10/36 (27.8%)	10/36 (27.8%)	14/36 (38.9%)	2/36 (5.6%)
Documented information is verified for accuracy during the MDTM	1/32 (3.1%)	7/32 (21.9%)	10/32 (31.3%)	12/32 (37.5%)	2/32 (6.3%)
Documented information is verified for accuracy after the MDTM	0/35 (0%)	9/35 (25.7%)	15/35 (42.9%)	9/35 (25.7%)	2/35 (5.7%)
Documented information is reviewed for accuracy only if ambiguity or misunderstanding arises during the course of treatment	2/35 (5.7%)	7/35 (20%)	15/35 (42.9%)	8/35 (22.9%)	3/35 (8.6%)
The quality of the MDTM record keeping at our institution is adequate	0/28 (0%)	1/28 (3.6%)	2/28 (7.1%)	17/28 (60.7%)	8/28 (28.6%)

Table 7
Participants' perceptions of MDTM organisation.

Participant ID	Comment on MDTM Organisation
ParticipantID 7	Conversations are often dominated by a few, more educated individuals and the multidisciplinary aspects are more of an illusion.
ParticipantID 14	MDTMs work well and are getting even better since guidelines are in place.
ParticipantID 15	Well-prepared cases, good summaries, and clear questions contribute to good discussions.
ParticipantID 16	Well-functioning leadership at the meetings is crucial for good discussions.
ParticipantID 17	Information about the patient's wishes is often missing.
ParticipantID 21	Organisation works well, but often technical turmoil delays the meetings and creates frustration for all participants.
ParticipantID 23	Usually, [there is] great common agreement in the decisions, but there is also the opportunity to present a different approach to the cases for discussion about the best treatment or care for the patient.
ParticipantID 24	A good working environment is extremely important; that the person handling the case is open to comments and perspectives from everyone in the room.
ParticipantID 29	There should be two specialists present in the current discipline, but often even more rewarding when more specialists are present
ParticipantID 34	It is better that the person presenting the patient knows the patient personally, but this is not always possible. The contact nurse can contribute with this information, alternatively that the contact nurse's notes should be attached in referrals.

applicable, iii) further referrals, iv) getting additional opinions, or in some cases v) discussing the various options with the patient. Reaching consensus in the MDTMs is important because the independent investigative processes involving different specialities, leading up to the MDTM, provide for a more reliable diagnosis than one made by any single specialist acting alone; apparent inconsistencies in the data are resolved through the meeting discussion to agree on the diagnosis [22]. Definitive diagnosis and disease staging are necessary to inform treatment and management decisions.

Insufficient information such as pathology results, information on co-morbidities, patient preferences, and lack of clinical data are quoted in literature as reasons for poor decision-making, inability to complete discussions, and recommendations not being implemented [38,39,2]. For our participants, clinical information and radiology contributions are mostly adequate, pathology contribution is sufficient in just a little over half of the cases, and co-morbid information and patient preferences have a lower adequacy rating. There is sometimes uncertainty regarding co-morbid information and patient preferences even though the discussions proceed and consensus is reached. Hahlweg et al. [35] report that if the case history and radiology information are of high quality, the likelihood of a recommendation being given increases.

Clinical practice guidelines (CPGs) are an important component of patient case discussions. Given that over 92% say that they either completely (13, 36.1%) influence the management/treatment recommendation or contribute to it to a great extent (22, 61.1%) demonstrates the important regard in which CPGs are held with respect to the treatment and management decisions. Clinical guidelines are central to the treatment and management recommendation and thus are likely to positively influence the outcome of the case discussion even in cases where co-morbid information and patient preferences may appear inadequate. Guidelines provide the rationale for specific treatment recommendations, and result in increased cooperation because the expectations of treatment are predefined, objective, based on scientific evidence, documented, and can be easily referenced [40]. In many cases, treatment recommendations may be agreed upon by presenting only medical information and not considering additional factors such as

patient views and psychosocial information which may lead to a limited view of the treatments most suitable for a patient case [35].

Documentation of the diagnosis and intended treatment is normal in medical practice; an accurate and complete record should exist for cases discussed at MDTMs. Accurate documentation aids the implementation of decisions taken at the MDTM [29]. Having documentation also ensures that there is consistency between written reports and opinions shared at the meeting [36].

Health professionals in the Kane and Luz [30] study identify the minimum data required in the MDTM records as the patient clinical summary, pathology and radiology opinion, diagnosis, TNM staging, treatment options, and the management plan. The TNM staging is the most widely used method of staging many cancers. The T refers to the size and extent of the main tumour; N refers to the number of lymph nodes affected and M refers to whether, or not, the cancer has metastasised [41].

Additionally, a summary of the discussion, names of pathologists and radiologists who gave opinions, and reasoning behind the decisions that are taken should, ideally, be included [30]. It is proposed that once the diagnosis, disease stage, and treatment plan are agreed upon, the decision should be clearly documented along with the rationale for the decision, and in some cases, the arguments leading to the decisions [30]. Making a clear distinction in the documentation between a generally recommended treatment according to guidelines and the proposed treatment for a specific case, taking into consideration co-morbidities and patient state, improves transparency [29].

Registering attendance in local MDTMs appears to be an established practice, despite that there are outliers where participants point out that the practice of noting attendees happens sometimes but not always, and that in some cases, not all attendees are recorded in the same manner. Attendance by key MDT members is important during the clinical decision-making process [42]. The documenting of attendance at meetings is necessary in order to be able to refer back to the relevant attendees should there be a need to clarify a matter, for further consultation in the event of similar patient cases, or even in order to identify responsible persons who are accountable for the decisions taken. Patient confidentiality is also better maintained through registering attendance as the lists can be used to ensure that only permitted attendees have access [43]. However, further information regarding latecomers and early leavers is necessary to avoid the incorrect reference to attendees in situations in which their absence applies [1]. National MDTMs in Sweden seem not to observe the same practices of taking attendance as local MDTMs. Further studies could elucidate reasons for the lack in taking attendance at the national level. The use of electronic systems could also aid the ability to easily access information on relevant attendees when contact is necessary.

Despite the MDT reaching an agreement, recommendations are not always followed. The reasons differ and are said to be justifiable in many cases. Ultimately the MDT recommendation is discussed with the patient, and the final decision on the treatment to be implemented is taken. Participants highlight slightly contradicting information in their text explanations to open-ended questions, see Table 8. Though these comments are given in various questions, they highlight that slight differences in the outlook towards patient preferences exist.

Table 8
Participant Comments on Patient Preferences.

Participant ID	Participant Comments on Patient Preferences
ParticipantID 7	This [consensus not being reached] is often due to a lack of information or knowledge of the patient's preferences. In such a case, the missing information is retrieved, and the case is discussed again.
ParticipantID 18	The point about the patient's preferences is overestimated regarding the outcome of the MDTM. As a rule, the patient may not have preferences that influence the decision on treatment. If the patient has views, it is usually that he or she does not want treatment at all.
ParticipantID 21	If the patient and the treating physician agree, MDTM recommendations can be deviated from [those agreed].

While ParticipantID 18 suggests a lack of flexibility towards the alteration of recommended treatment by patient preferences, ParticipantID 7 suggests that a patient's preferences play an important role in finalizing the treatment decision. ParticipantID 21, on the other hand, implies that patients, along with their doctors, may prefer an alternative treatment to that which has been recommended by the MDT. It must be noted, however, that scenarios and motivating factors that drive these situations may differ. According to Rajan et al. [44], the alterations to MDT recommendations are usually due to patient choice or clinical information that had not previously been considered. Deviations must be recorded [30], regardless of the reason for not adhering to recommendations. This practice aids transparency, ensuring there is no confusion for other treating professionals arising from a discrepancy between the MDT record recommendation and the treatment the patient receives. The record of reasons for deviation will also be an opportunity for further learning on why certain recommendations may be considered unsuitable despite the MDT agreement.

The discussion of their case at an MDTM gives patients the opportunity for evaluation by the team of specialists, thus providing a higher probability that the various aspects of their health and different treatment options are thoroughly considered. The MDT has a substantial influence on treatment decisions [45], and adherence to treatment recommendations is higher for cases discussed by the MDT [46–48]. As such, statistically significant differences are seen in survival rate for stage-IV head and neck cancer patients managed by MDTs [45], and lower distant recurrence and mortality rates for breast cancer patients who comply with MDTM recommendations [48]. However, inclusion of patients in MDTM decision-making, and considering patient preferences to ensure that decisions align with patient wishes [49], and co-morbidities to ensure recommended treatments are well suited for their condition [50], will positively influence adherence rates. Documentation is a way to keep track of MDT recommendations and can be used to perform a comparative analysis between patients whose treatment is compliant versus non-compliant [51]. A comprehensive MDTM record that encompasses all important aspects of the discussion and decision-making, and the sharing of records, empowers both the healthcare provider and patients. A study on breast cancer patients highlights that information sharing and being discussed at MDTMs improves patients' perceptions of care [52], and access to electronic medical records by patients is shown to help patients feel more in control of their care [53].

5.2. Current record-keeping practices

Record-keeping is conducted electronically, via taking personal notes and filling forms or paper records. Several participants highlight that notes are later dictated to someone who enters them into the electronic medical system, while in some cases the note taker may update the record on the electronic medical record system. It is reasonable to expect that dictating personal notes is less likely to result in errors introduced by poor handwriting or the use of shorthand, abbreviations, and symbols [31]. For most participants, record capturing happens post-meeting (as soon as MDTM is over). Bates et al. [54] discuss that electronic records are beneficial in that they improve legibility, completeness, organisation of information, as well as provide decision support. Real-time electronic documentation during the meeting is considered best [44,22], but it has also been met with challenges. Improvements in real-time data collection and feedback could potentially improve quality, care coordination and patient-centered models of care [55]. Real-time note-taking would also remove duplication of documentation work [23].

Over half of the participants (18/32, 56.3%) report that the responsibility for documenting MDTM proceedings with regard to patient

case discussions lies mostly with the doctor who referred the patient case to the MDTM. However, a long list of alternatives or supplementary roles is given by participants (Table 3). Several concerns may arise from the mentioned mixed practice. If there is no standardized format, the quality of documentation will differ based on the person who has been tasked to document the proceedings. Several doctors refer patient cases to the MDT for discussion. While those referring (attending) doctors are responsible for documentation, several others also take notes at the MDTM. A shared electronic medical record system to which all information and notes are recorded can be expected to bring structure to MDTM records [56]. Shared electronic medical records aim at aiding cooperation and coherence and increasing the quality of medical treatment and care in a multidisciplinary setting [57]. Christiansen et al. [57] discuss that in a shared record system used for ulcer treatment, all collaborating health workers are expected to update in the same recording system and not in their respective internal health record systems to avoid dual documentation. The data collected should be in a format that is usable and meaningful to the clinician.

The chairman is important to the running of MDTMs [58] and maintaining order as this ensures better discussions and better likelihood for a discussion to reach the desired conclusion. In the study by Lamb et al. [58] observers witnessed that unsystematic presentation of information and contribution to the MDTM discussion made data capture more difficult. Similarly, MDTM documentation would benefit from structured discussions. Because the chairman is a medical professional who needs to express her/his views while maintaining meeting order and recording information to be later dictated and entered into the record system, it is possible for some information to be missed in the documentation process. According to Bothin and Clough [59], people take more notes while they are engaged in discussions. But the number of discussed cases in MDTMs may be overwhelmingly large. Therefore, the question arises as to whether the chairman is best suited to handle the documentation responsibility as well as the other meeting functions simultaneously.

With personal notes, various MDT members record different information according to its relevance to their interests. People take personal notes to keep a record of vital issues and to help them remember details discussed, but some rely on the confidence that as experts in an area, they will not forget [59]. In their experiments, Bothin and Clough [59] found in the different experimental settings that short and frequent notes are taken, salient items are more likely to be noted in personal notes, different people take notes at different times during the meeting, and people often note the key summary items such as decisions and action items. Bothin and Clough's [59] experiments also show that people take notes of about 50% of the salient information otherwise recorded in the reference summary. While there is no guarantee that all information is captured in MDTM personal notes, the information contained in personal notes is a good indication of the details that should be contained in the MDT medical record and information systems designed for record-keeping. A general record of events in a meeting may suffice for those who attend meetings; for the purpose of updating non-attendees, however, detailed records are necessary. Another area of concern is that personal notes may not truly represent what was said at the meeting due to misunderstanding, and thus there is no guarantee that the person making an utterance is not misquoted in the future [7]. It can be argued that personal notes can be useful to the individual taking the note but are risky from an organisational viewpoint as it is not feasible to validate personal notes. Having a validated meeting record that contains all the information needed by MDT members is the ideal.

Comments made allude to the practice that different specialists record in different electronic systems, i.e. pathologists, radiologists, and surgeons. It is unclear whether these systems converge or whether they are all interlinked with the electronic system where MDTM records are stored. The differences highlighted in this study show that there is little uniformity with regard to how documentation is carried out, and neither is it clear about the professional role that is responsible for either the meeting record (or minutes) or the record for the patient file. For example, a consultant surgeon or physician may lead the MDT and may also be the professional role responsible for the record [7]. It is not unusual for a medical registrar (attending physician) or specialist nurse to write the record of the discussion. There is no clear standard regarding the registration of attendance of MDT members at meetings, and the task of noting attendance is often assigned to a clerical staff member [7].

6. Future Directions

If there were a comprehensive record of patient case discussions at MDT meeting, opportunities would be generated for the application of novel technologies that could make the MDT and patient care processes more effective and efficient. Advances in the application of artificial intelligence (AI) methods in natural language processing (NLP) suggests that semantics can be extracted from the discussion to make the content available in an understandable way to all participants [60,61]. Indeed recent analysis of online discussions in the context of the Covid-19 pandemic prompts the potential for automated extraction discussion topics, the use of long short-term memory (LSTM) recurrent neural network for sentiment classification and a natural language process (NLP) method based on topic modeling to uncover various issues of interest [62], could be potentially investigated for use with MDTMs.

Although there are challenges around capturing multimodal interaction at MDTMs, the potential benefit that such meeting records promise include automatic data collection for national statistics, and facilitate real time review of similar case to current discussion [63]. A system capable of automating the collection of content-free (paralinguistic) features and segmenting the recorded data into meaningful subunits would provide a tool for analyzing meetings with respect to their effectiveness, the impact of new meeting-support technologies on the interaction, as well as facilitate the employment of AI methods [64]. Systems that can unify access to heterogenous data sources such as EHR, national and international data sources, and medical literature databases might enhance the educational and professional development roles of the MDTM as well as enable participants to further enrich the case discussions [65].

In this study we enquired generally about record-keeping practices at MDTMs. We did not ask for more detailed information about the extent to which participants use information from structured data entry (SDE) forms, the standards that are used, such as ICD, ATC, SNOMED, etc. Further study directed at medical informaticians to gather quantitive data in this respect would be a great help to developers.

Research on the element of patient-centeredness in multidisciplinary medical teamwork, and MDTMs is lacking. There are anecdotal reports of head and neck cancer patients in attendance at the MDTMs when their case is being discussed in some teaching hospital settings in UK, USA and Australia, but there is little documented record or research in this area. While MDTMs foster adherence to clinical practice guidelines, research tends to point to clinician involvement in the MDTM helping patients in their adherence, rather than direct involvement of patients in the decision-making process [35]. More research is needed to establish practices and experiences around MDTM decisions and patient involvement in MDTMs. From a patient perspective, research should investigate mechanisms so that patient co-morbidities and preferences are sufficiently considered when treatment recommendations are being determined. From an organisational perspective, research should examine how best to implement MDTM recommendations and identify deviations from recommended clinical care pathways, and practice guidelines.

A comprehensive electronic record system would allow for algorithms to be developed that could automatically link pathology, clinical and radiological coding to diagnostic codes and clinical practice guidelines and thus alleviate some cases from the burgeoning lists of cases being currently identified for MDT discussion [8]. Automatically,

or semi-automatically, selected cases could be validated by new processes within the MDT structures.

7. Conclusion

Our study focuses on multidisciplinary team management of cancer cases, however, multidisciplinary care applies also to other complex medical conditions such as diabetes [66], pain management [67], and major burns [68]. The concept of a structured, standardized, and comprehensive MDTM record also applies in these areas to aid good communication and transparency.

We show that the reported practices of MDTs appear to be aligned with the processes described in literature, and suggest that there is scope for better reconciliation between the patient preferences and clinical practice guidelines in the treatment recommendations agreed.

Record-keeping is reported by our participants of this study to be of satisfactory quality: the required information is present in the record, and a high level of satisfaction is expressed. Records are mostly stored electronically and used by most of the participants for reference during patient care, which is a commendable step for MDTM record-keeping as electronic records increase possibilities of completeness and increased accessibility.

However, the results highlight the need for a standardised record-keeping process in MDTMs and ensuring a centralised repository that captures the important components of the discussion and decision-making, as well as the tasks, roles, and treatment intent for each patient case. In spite of the said quality in records, there are many different parties responsible for record-keeping, and several different systems may be in use for documenting parts of the discussion and decisions by different specialities. There is still a need for further research regarding the necessity and the effect of having many different parties responsible for record-keeping. There is the question of whether information captured by various roles in a record satisfies the needs of all MDT members, as well as whether or not it is available in a shared record. Some uncertainties exist regarding record verification and validation, and expectations on capturing attendance lists ought to be clarified. Problems with technical equipment at MDTMs are raised, which threatens both the quality of the meeting and the related record-keeping.

The concerns identified appear to undermine the high satisfaction levels reported on the adequacy of MDT record quality. Further research should elucidate on the concerns raised.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

We wish to thank those who kindly gave their experience in the research survey.

Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at https://doi.org/10.1016/j.ijmedinf.2022.104711.

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